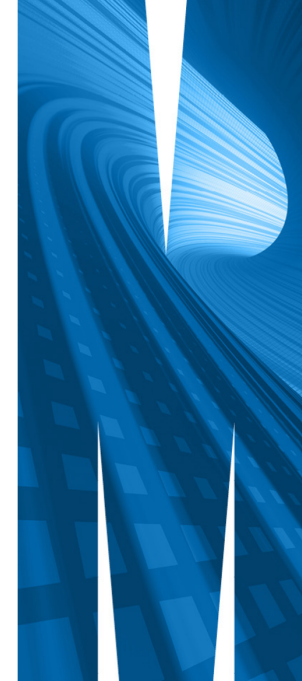
MONASH University

ETX1100/ETX5900 Business Statistics

Week 4: Testing business claims

15 December 2026



Outline

1. From a claim to a decision
2. Testing one population mean
3. Testing a population proportion
4. Comparing two independent means
5. Decisions and their consequences

Today's question

When does a sample provide enough evidence to challenge a business claim?

Follow one ETF lecture sequence: claim $\rightarrow$ hypotheses $\rightarrow$ sampling model $\rightarrow$ test statistic $\rightarrow$ p-value $\rightarrow$ decision $\rightarrow$ business meaning. Apply it to one mean, a proportion, and a difference between two independent means.

Outline

1 From a claim to a decision

2 Testing one population mean

3 Testing a population proportion

4 Comparing two independent means

5 Decisions and their consequences

Why test a hypothesis?

A sample estimate rarely equals a stated benchmark exactly. A test asks whether the difference is **too large to explain readily by sampling variation**, assuming the benchmark is true.

- The **null hypothesis** states the benchmark.
- The **alternative hypothesis** states the direction of evidence sought.
- The **significance level** α is the preselected threshold for a false rejection.
- The **p-value** measures how extreme the observed statistic is under the null model.

State the hypotheses before looking

Business claim	Null	Alternative	Tail
Average wait exceeds 30 minutes	$H_0 : \mu = 30$	$H_1 : \mu > 30$	right
Mean sale price differs from \$1.2m	$H_0 : \mu = 1.2$	$H_1 : \mu \neq 1.2$	two
Fewer than 5% of invoices have	$H_0 : \pi = 0.05$	$H_1 : \pi < 0.05$	left

Watch me: click-through claim

ETF's workshop asks whether a population click-through mean is **below 12 clicks per minute**. Before using the data, write $H_0 : \mu = 12$ and $H_1 : \mu < 12$.

The word *below* specifies the left tail. The observed sample mean determines the statistic, but it must not determine which alternative we choose.

Now you try: translate four claims

Time allowed: 5 minutes

Write the parameter, H_0, H_1 and tail for: average delivery time above 30 minutes; mean price different from \$1.2m; invoice error rate below 5%; and more than half of customers using an app.

The five-step testing workflow

1. Define the population parameter and write H_0 and H_1.
2. Check the sampling design and assumptions; calculate the statistic.
3. Calculate the p-value in the tail specified by H_1.
4. Compare p-value with the chosen α.
5. Report a contextual conclusion, effect size and important limitations.

The same order works for a mean, a proportion and a difference.

A right-tail p-value is an area

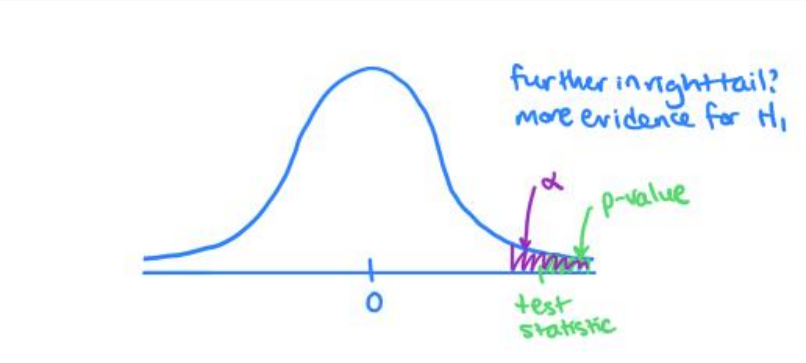


Figure 1: ETF Seminar Week 6: the p-value is the tail area at least as extreme as the observed statistic, assuming the null model.

For a left-sided claim, use the left tail; for a two-sided claim,

Outline

1 From a claim to a decision

2 Testing one population mean

3 Testing a population proportion

4 Comparing two independent means

5 Decisions and their consequences

From estimate to t statistic

When the population SD is unknown, use the sample SD:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1.$$

The numerator is the observed difference from the null benchmark. The denominator is the standard error of the sample mean. Check independence and whether the t approximation is reasonable for the data.

Excel: choose the correct tail

Alternative	Excel formula
$\mu < \mu_0$	`=T.DIST(t,df,TRUE)`
$\mu > \mu_0$	`=1-T.DIST(t,df,TRUE)`
$\mu \neq \mu_0$	`=T.DIST.2T(ABS(t),df)`

Keep the signed statistic visible. `ABS` belongs only in the two-sided tail formula.

Watch me: ETF click-through example

A random sample has n=45, $\bar{x}$ = 10.8 and s=3.2 clicks per minute. Test $H_1 : \mu < 12$ at α = 0.05.

1. Put the inputs in labelled Excel cells.
2. Compute `=(10.8-12)/(3.2/SQRT(45))`.
3. Use `=T.DIST(t_cell,44,TRUE)`.
4. Compare the p-value with 0.05 and state the conclusion in clicks per minute.

ETF Workshop Week 6 supplies the business context; keep your worksheet formulas auditable.

Now you try: student spending

Time allowed: 8 minutes

A random sample of 40 students has mean monthly discretionary spending $283 and SD $17.85. Test at 5% whether the population mean **exceeds $275**. Show the five steps and an Excel p-value formula. State what the sample does and does not establish.

One-sided and two-sided are different questions

A two-sided alternative asks whether the population mean **differs** in either direction. The statistic stays the same, but the p-value includes both tails.

Choose the question before inspecting the result. Changing the tail to obtain a smaller p-value invalidates the planned error rate.

Now you try: change only the claim

Time allowed: 5 minutes

Revisit the spending sample with $H_1 : \mu \neq 275$. Identify which entries in your worksheet change and compare the two conclusions.

Outline

1 From a claim to a decision

2 Testing one population mean

3 Testing a population proportion

4 Comparing two independent means

5 Decisions and their consequences

Count an event, then test a proportion

For $H_0 : \pi = \pi_0$, first count successes x from n observations and calculate $\hat{p} = x/n$. The large-sample test statistic is

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\pi_0(1 - \pi_0)/n}}.$$

Check $n\pi_0 \geq 5$ and $n(1 - \pi_0) \geq 5$. A **test** uses the null value π_0 in its standard error; an interval uses $\hat{p}$.

Excel: Normal tail areas

Alternative	Excel p-value
$\pi < \pi_0$	`=NORM.S.DIST(z,TRUE)`
$\pi > \pi_0$	`=1-NORM.S.DIST(z,TRUE)`
$\pi \neq \pi_0$	`=2*(1-NORM.S.DIST(ABS(z),TRUE))`

Use a clearly labelled success count and denominator. Check that the selected tail agrees with the claim.

Watch me: ETF bank transaction example

ETF's workshop gives a 95% error-free benchmark and **229 successes among 250** randomly sampled transactions.

Test $H_0 : \pi = 0.95$ against $H_1 : \pi < 0.95$ at $\alpha = 0.10$. In Excel, calculate $\hat{p}$, the two null expected counts, z and the left-tail p-value. Interpret the decision as evidence about the bank's population success rate.

Now you try: invoice accuracy

Time allowed: 8 minutes

An audit finds 288 accurate invoices among 300. Test at 5% whether the population accuracy rate is **below 98%**. Show the count check, statistic, Excel formula, decision and one-sentence business conclusion.

Outline

1 From a claim to a decision

2 Testing one population mean

3 Testing a population proportion

4 Comparing two independent means

5 Decisions and their consequences

A difference has its own question

Suppose we compare population mean sale prices for properties **inside** and **outside** a school zone. Define μ_Y and μ_N for the two groups and estimate

$$\hat{\Delta} = \bar{x}_Y - \bar{x}_N.$$

For a directional claim that school-zone prices are higher, use $H_0 : \mu_Y - \mu_N = 0$ and $H_1 : \mu_Y - \mu_N > 0$. The order of subtraction controls the tail.

Why the two-sample standard error is different

Two independent groups contribute uncertainty:

$$SE(\bar{x}_Y - \bar{x}_N) = \sqrt{\frac{s_Y^2}{n_Y} + \frac{s_N^2}{n_N}}.$$

The Welch statistic is $t = (\hat{\Delta} - 0)/SE$. Its degrees of freedom account for both group sizes and variances; Excel's unequal-variance ToolPak test calculates them. Independence matters **within and between** groups.

Watch me: prepare the RealEstate comparison

Open the **RealEstate** sheet of `week4-proportions-data.xlsx`.

1. Identify `Price` and `SchoolZone`, their units and any missing rows.
2. Filter `SchoolZone=Yes` and `SchoolZone=No`; copy each group's prices to a separate labelled column on a working sheet.
3. Use `COUNT`, `AVERAGE` and `STDEV.S` on each price column.
4. Calculate $\bar{x}_Y - \bar{x}_N$ and the Welch standard error.
5. Compare the difference with a business-relevant dollar threshold before discussing the test.

Excel ToolPak: unequal variances

Check **Data** $\rightarrow$ **Data Analysis** is available. Choose **t-Test: Two-Sample Assuming Unequal Variances**.

- Select the two price columns as Variable 1 and Variable 2, with labels if included.
- Enter **0** for hypothesised mean difference and the preselected α.
- Read the group means, `t` statistic, `df` and one-tail p-value.
- Match Variable 1 minus Variable 2 to your alternative; the ToolPak's one-tail field must correspond to the observed direction.

Now you try: run and audit Welch

Time allowed: 12 minutes

Using the RealEstate workbook, test whether mean school-zone price **exceeds** mean non-school-zone price at 5%.

Report `n`, mean and SD for each group; the mean difference and its units; Welch `t`, `df` and one-tail p-value; and a plain-language conclusion. Confirm that the ToolPak groups are in the order named in your hypotheses.

Independent groups and a paired design

The school-zone groups contain different properties, so the independent two-sample test is appropriate. When the **same units** are measured before and after a change, analyse their within-unit differences instead.

Do not pair unrelated rows merely because they appear on the same line of a spreadsheet.

Outline

1 From a claim to a decision

2 Testing one population mean

3 Testing a population proportion

4 Comparing two independent means

5 Decisions and their consequences

What a decision can say

If $p < \alpha$, **reject** H_0 and describe the evidence for H_1. If $p \geq \alpha$, **fail to reject** H_0: the sample does not give sufficient evidence for H_1 at that threshold.

Failure to reject does not prove equality. The p-value does not measure the chance that the null is true.

Type I and Type II errors

Decision	H_0 true	H_0 false
Reject H_0	Type I error	correct rejection
Fail to reject H_0	correct non-rejection	Type II error

A Type I error has probability α when the null assumptions hold. Reducing α makes rejection harder and can increase Type II risk when other factors remain fixed.

Watch me: name errors in context

For the bank test, a **Type I error** means concluding that population transaction success is below 95% when it is not. A **Type II error** means missing a real shortfall below 95%.

Explain whose decision changes if either error occurs. The costs of the errors help select α before viewing the sample.

Now you try: business consequences

Time allowed: 6 minutes

For the invoice-accuracy test, describe both errors in everyday language. Which error would be more costly to an operations manager, and why? How would that judgment affect the chosen significance level?

Statistical versus practical significance

A small p-value concerns **evidence against the null**. It does not say the effect is large enough to matter.

Report the observed difference, its units, a confidence interval when appropriate, and a business threshold. For two groups, ask whether a plausible difference is large enough to change a decision. A large sample can make a small difference statistically significant.

Now you try: challenge the headline

Time allowed: 5 minutes

A report says, “Waiting time fell by 0.2 minutes, $p < 0.001$, so the redesign was a major success.” Identify what the p-value supports, what further information you need, and a more defensible headline.

A1 preparation: keep the measures separate

The current assignment asks you to distinguish four different outcomes: **dropped calls**, **support reports**, **all-cause returns**, and **measured handset reception loss**. They have different units and denominators.

For the independent two-sample mean test, define the direction of the reception-loss difference, use the unequal-variance method, then compare statistical evidence with the stated **3 dB** practical threshold. A business conclusion also needs sampling and measurement caveats.

Final audit of an Excel testing worksheet

Can another person trace the population and event, raw rows, group counts, inputs, assumptions, hypotheses, formula or ToolPak command, tail, decision and conclusion?

Claim $\rightarrow$ **method** $\rightarrow$ **Excel execution** $\rightarrow$ **validation** $\rightarrow$ **plain-language conclusion.** Keep the effect size and practical threshold visible beside the p-value.